Yousef Fahim Amin Hassoun

Final-Year Computer Science Undergraduate | Machine Learning & Computer Vision

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ACADEMIC PROFILE

Final-year Computer Science undergraduate seeking master's study in Computer Science and Artificial Intelligence. Project experience in machine learning, real-time computer vision, and database-backed web applications. Academic interests include applied machine learning, model evaluation, and interactive vision systems.

EDUCATION

B.Sc. in Computer Science

2023 - 2027 (expected)

Modern Academy for Computer Science and Management Technology, Maadi, Cairo, Egypt

Current cumulative GPA: 2.61/4.0 | **Official classification:** Very good-
Language of instruction: English
Relevant mathematics coursework: Linear Algebra, Calculus, Probability & Statistics

SELECTED PROJECTS

airDrums | Computer Vision & Motion Prediction | GitHub

Python, OpenCV, MediaPipe, PyTorch, LightGBM, OpenGL, sounddevice, pytest

- Developed an offline webcam-based drumming application with fingertip and drumstick tracking, multiple kit layouts including a seven-piece kit, and a GPU-rendered augmented reality view.
- Integrated motion-based strike detection, duplicate-hit suppression, calibration, and recorded-sample audio playback.
- Implemented GBDT, GRU, and TCN models for offline motion-prediction experiments, with training and evaluation pipelines using synthetic and developer data.

HeartGuard AI | Heart Disease Classification | GitHub

Python, scikit-learn, Pandas, NumPy, Streamlit, Matplotlib

- Built a machine learning pipeline using the UCI Heart Disease dataset to compare classification models and feature subsets.
- Explored PCA and recursive feature elimination (RFE); developed a Streamlit interface to display validation metrics and generate model predictions.

RealEstate | Property & Lead Management Platform | GitHub

Python, Flask, MySQL, Jinja2, JavaScript, pytest

- Built a relational web application connecting property listings, agents, client inquiries, and analytical dashboards.
- Implemented referential integrity, parameterized SQL, lead status transitions, and aggregate queries for dashboard indicators.

TECHNICAL SKILLS

Machine learning & vision
scikit-learn, XGBoost, TensorFlow, Pandas, NumPy, OpenCV, MediaPipe, PCA, RFE

Software & databases
Flask, MySQL, SQL, Streamlit, Git/GitHub, pytest

Programming
Python, C, C++, Java, JavaScript, HTML/CSS; Embedded C and Arduino

TRAINING & CERTIFICATES

Full Stack with Python
Ray Group (45 days)

AI & Machine Learning
Sprints x Microsoft Summer Camp Certificate

AI & Machine Learning
IT Gate Academy, Nasr City (40 hours)

Python Programming
Ex-Fresher Certificate

LANGUAGES

Arabic: Native | **English:** Proficient (academic and technical communication)

DECLARATION & DATA PRIVACY AUTHORIZATION

I hereby authorize the processing of my personal data contained in this Curriculum Vitae in compliance with the European Union General Data Protection Regulation (EU GDPR 2016/679) and the Italian Legislative Decree No. 196/2003 (as amended by Legislative Decree No. 101/2018) for academic admission and evaluation purposes.